

1. Import the [data table Download data table](#) and look at it. Do you see any issues you should deal with? (HINT: there are some missing values)

Some of the countries years have data missing as N/A

2. Create a histogram and a smoothed density estimate of the 2006, 2010, and 2013 scores.

The screenshot shows an RStudio interface with a script editor on the left, a data viewer on the right, and a console at the bottom.

Script Editor:

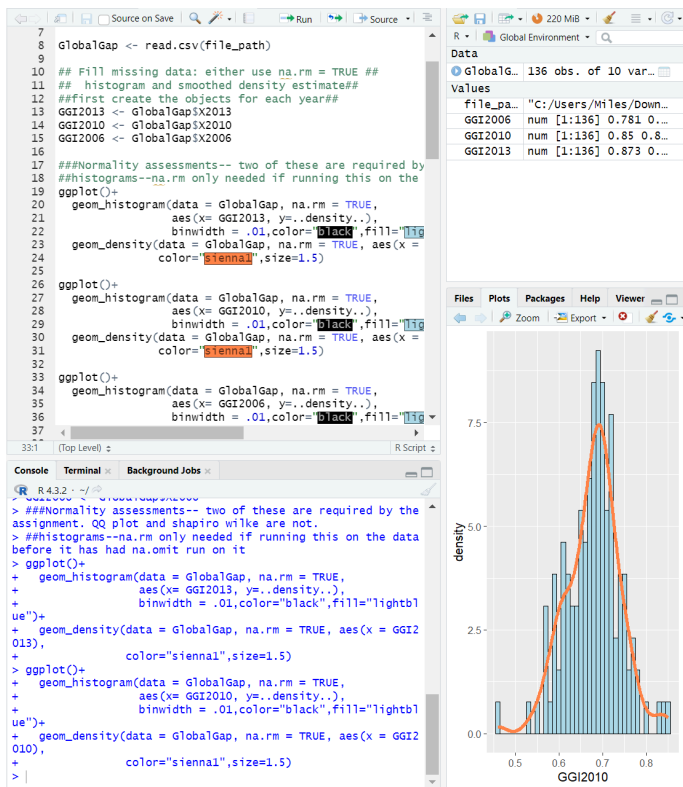
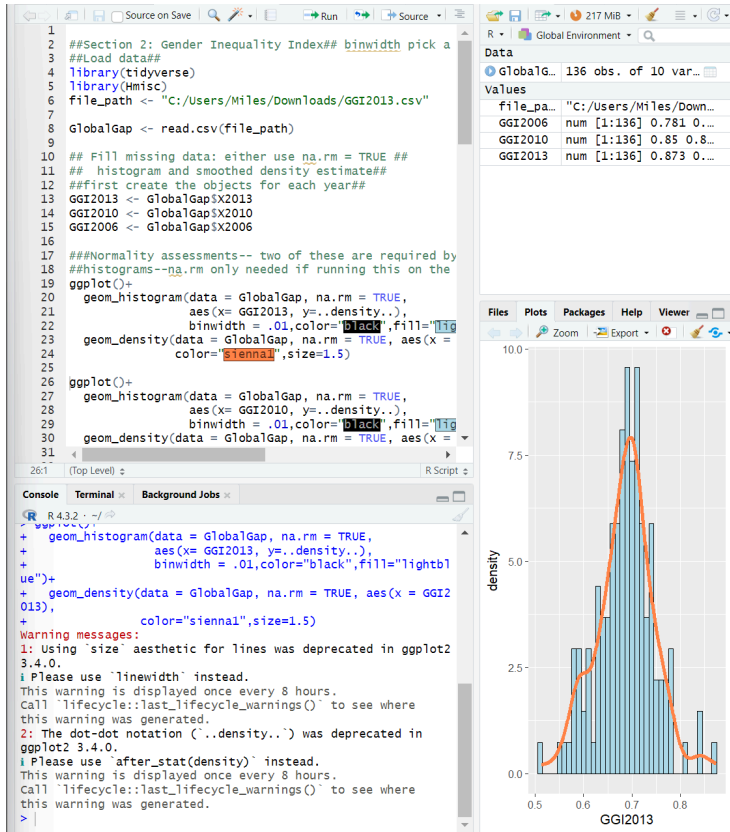
```
1  
2 ##Section 2: Gender Inequality Index## binwidth pick a  
3 ##Load data##  
4 library(tidyverse)  
5 library(Hmisc)  
6 file_path <- "C:/Users/Miles/Downloads/GGI2013.csv"  
7  
8 GlobalGap <- read.csv(file_path)  
9  
10 ## Fill missing data: either use na.rm = TRUE ##  
11 ## histogram and smoothed density estimate##  
12 ##first create the objects for each year##  
13 GGI2013 <- GlobalGap$X2013  
14 GGI2010 <- GlobalGap$X2010  
15 GGI2006 <- GlobalGap$X2006  
16  
17 ###Normality assessments-- two of these are required by  
18 ##histograms--na.rm only needed if running this on the  
19 ggplot()+  
20   geom_histogram(data = GlobalGap, na.rm = TRUE,  
21                 aes(x= GGI2013, y=..density..),  
22                 binwidth = .01,color="black",fill="light  
23   geom_density(data = GlobalGap, na.rm = TRUE, aes(x =  
24               color="sienna",size=1.5)  
25  
26 ggplot()+  
27   geom_histogram(data = GlobalGap, na.rm = TRUE,  
28                 aes(x= GGI2010, y=..density..),  
29                 binwidth = .01,color="black",fill="light  
30   geom_density(data = GlobalGap, na.rm = TRUE, aes(x =  
31
```

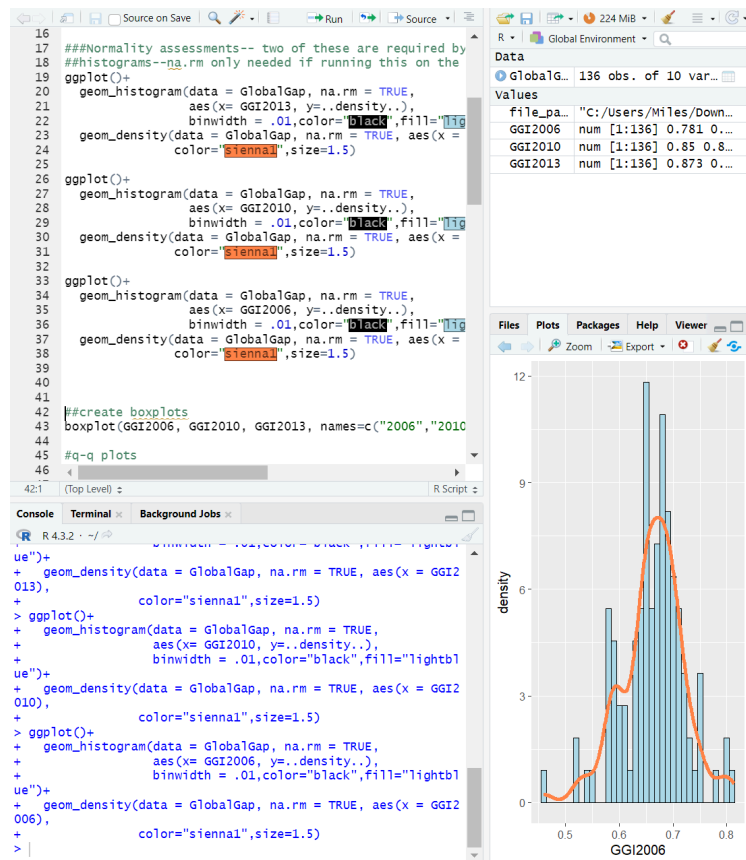
Data Viewer:

GlobalG...	
file_pa...	"C:/Users/Miles/Down...
GGI2006	num [1:136] 0.781 0...
GGI2010	num [1:136] 0.85 0.8...
GGI2013	num [1:136] 0.873 0...

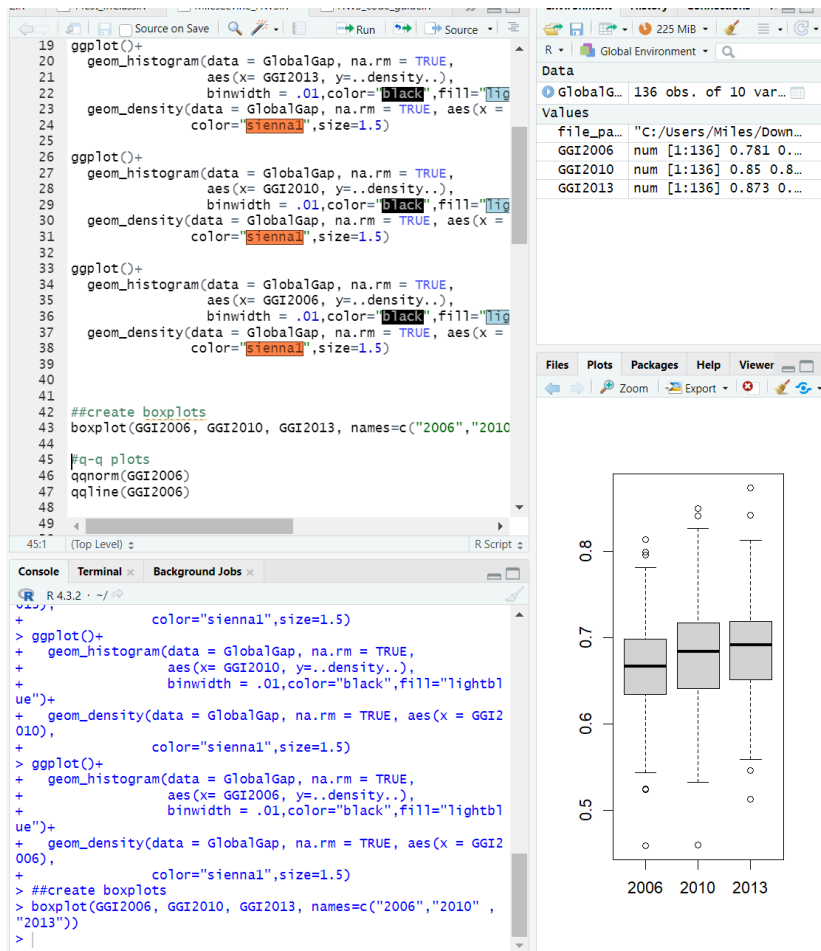
Console:

```
R 4.3.2 ~/  
> ##Section 2: Gender Inequality Index## binwidth pick a bindwit  
> that will divide your data so like .01  
> ##Load data##  
> library(tidyverse)  
> library(Hmisc)  
> file_path <- "C:/Users/Miles/Downloads/GGI2013.csv"  
> GlobalGap <- read.csv(file_path)  
> ## Fill missing data: either use na.rm = TRUE ##  
> ## histogram and smoothed density estimate##  
> ##first create the objects for each year##  
> GGI2013 <- GlobalGap$X2013  
> GGI2010 <- GlobalGap$X2010  
> GGI2006 <- GlobalGap$X2006  
>
```

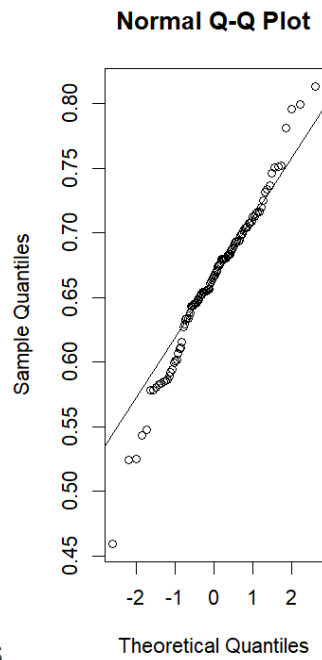




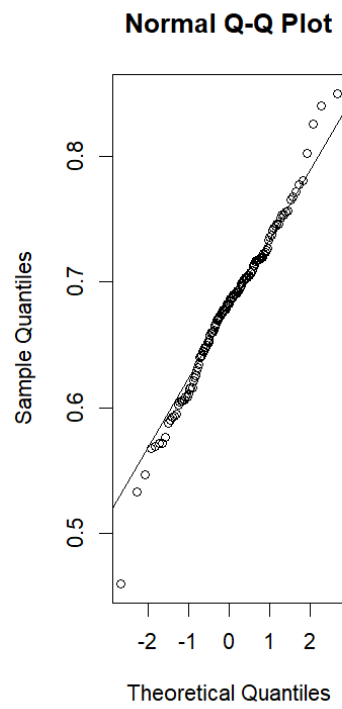
3. Next create a box and whisker plot of the same data.

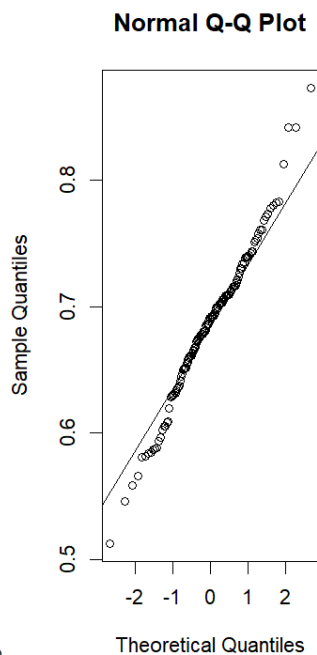


2006



2010





2013

Shapiro-Wilk normality test

data: GGI2006

W = 0.97782, p-value = 0.06336

```
> shapiro.test(GGI2010)
```

Shapiro-Wilk normality test

data: GGI2010

W = 0.98605, p-value = 0.2075

```
> shapiro.test(GGI2013)
```

Shapiro-Wilk normality test

data: GGI2013

W = 0.98719, p-value = 0.2384

```
> |
```

4. Calculate summary data for each year you have plotted. Have the means changed over time? How about the standard deviations?

```

> #display the central tendency for each year
> summary(GGI2006)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
0.4595 0.6346 0.6667 0.6626 0.6973 0.8133    26
> summary(GGI2010)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
0.4603 0.6422 0.6839 0.6785 0.7167 0.8496     6
> summary(GGI2013)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.5128 0.6517 0.6913 0.6864 0.7177 0.8731
> mean_cl_normal(GGI2006)
      y      ymin      ymax
1 0.66264 0.6515759 0.6737041
> mean_cl_normal(GGI2010)
      y      ymin      ymax
1 0.67855 0.6677909 0.6893091
> mean_cl_normal(GGI2013)
      y      ymin      ymax
1 0.6863941 0.6761819 0.6966063

```

5. Do the data look normally distributed based on what you know thus far?

Yes this data looks normally distributed because each year has similar values for min, 1st Qu, median, mean, 3rd Qu and max.

6. Calculate the z-scores for each of the years (2006, 2010, 2013).

```

> #2006 z-score vector#
> z_scoresGGI2006 <- (GGI2006-mean(GGI2006))/sd(GGI2006)
> #2010 z-score vector#
> z_scoresGGI2010 <- (GGI2010-mean(GGI2010))/sd(GGI2010)
> #2013 z-score vector#
> z_scoresGGI2013 <- (GGI2013-mean(GGI2013))/sd(GGI2013)
> print(z_scoresGGI2006)
 [1] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[25] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[49] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[73] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[97] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[121] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

```

```

> print(z_scoresGGI2010)
[1] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[25] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[49] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[73] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[97] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
[121] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA

> print(z_scoresGGI2013)
[1] 3.100481572 2.585688318 2.579045824 2.100786285 1.607581135
[6] 1.592635524 1.552780563 1.519568095 1.448161289 1.413288197
[11] 1.361808872 1.238922740 1.235601493 1.194085908 1.122679102
[16] 1.106072868 1.072860400 0.956616762 0.951634892 0.931707411
[21] 0.906798060 0.885209956 0.876906839 0.873585592 0.871924969
[26] 0.805500033 0.790554422 0.737414473 0.725790110 0.667668291
[31] 0.626152706 0.587958367 0.564709640 0.549764029 0.509909068
[36] 0.501605951 0.498284704 0.483339093 0.456769119 0.438502261
[41] 0.428538521 0.418574781 0.387022936 0.382041066 0.373737949
[46] 0.368756079 0.367095455 0.367095455 0.340525481 0.325579870
[51] 0.318937377 0.287385532 0.279082415 0.277421792 0.257494311
[56] 0.252512441 0.247530570 0.225942466 0.220960596 0.214318102
[61] 0.197711868 0.141250673 0.139590049 0.118001945 0.106377581
[66] 0.106377581 0.098074464 0.088110724 0.073165113 0.073165113
[71] 0.034970775 0.005079554 -0.001562940 -0.011526680 -0.026472291
[76] -0.087915356 -0.101200344 -0.101200344 -0.104521590 -0.127770318
[81] -0.136073435 -0.151019046 -0.156000916 -0.170946526 -0.185892137
[86] -0.189213384 -0.202498371 -0.222425852 -0.232389592 -0.302135775
[91] -0.322063256 -0.340330113 -0.353615101 -0.381845698 -0.416718790
[96] -0.423361283 -0.431664400 -0.439967517 -0.468198115 -0.504731830
[101] -0.519677441 -0.574478013 -0.582781130 -0.589423623 -0.607690481
[106] -0.655848559 -0.689061027 -0.750504093 -0.816929029 -0.821910899
[111] -0.851802121 -0.880032718 -0.916566433 -0.929851420 -0.938154537
[116] -0.949778901 -0.959742642 -1.105877501 -1.280242958 -1.300170439
[121] -1.346667894 -1.346667894 -1.388183479 -1.491142130 -1.542621455
[126] -1.625652625 -1.635616365 -1.647240729 -1.692077561 -1.697059431
[131] -1.743556886 -1.750199380 -1.997632267 -2.118857775 -2.333078194
[136] -2.882744539

```

7. Create a confidence interval for the mean for each of the years (2006, 2010, 2013).

```

> mean_cl_normal(GGI2006)
      y      ymin      ymax
1 0.66264 0.6515759 0.6737041
> mean_cl_normal(GGI2010)
      y      ymin      ymax
1 0.67855 0.6677909 0.6893091
> mean_cl_normal(GGI2013)
      y      ymin      ymax
1 0.6863941 0.6761819 0.6966063

```

8. Re-Import the data and remove the rows with missing data using na.omit if you haven't already. New z-scores for 2006 and 2010:

```

> print(z_scoresGGI2006)
[1] 2.02670005 2.27435850 2.33584611 2.57325662 1.51942724 1.21028119 1.50747132
[8] 1.42719582 0.63298082 -0.10316255 0.77132795 0.79353181 1.06510210 1.53309116
[15] 0.30846286 0.85160344 1.26152087 0.61419294 0.91992302 0.07617632 0.70984034
[22] 0.91650704 -0.33032512 -0.49770807 0.76961996 1.18295336 0.52879347 0.51512956
[29] 0.33237471 0.34603862 0.72179626 0.29138297 0.52708549 0.20256753 -0.32349317
[36] 0.41606618 0.40581825 -0.18173006 0.29138297 0.66201664 0.88576323 0.06422040
[43] 0.50488163 0.85672741 0.44851798 0.29992291 0.97799465 -0.41230861 0.60906897
[50] -0.13049038 0.54245739 0.24526726 -0.14244630 0.19744356 0.29138297 0.34945460
[57] 0.70300838 -0.28079343 -0.11170250 0.29138297 -0.29104137 0.02152066 0.22306340
[64] -0.60872737 0.04543251 -0.13219837 -0.23980169 -0.33544909 -0.01263912 -0.14757027
[71] -0.24492566 0.14620388 -0.18514604 0.46218189 0.12570801 0.12229203 -0.12024244
[78] -0.29274936 -0.14586228 0.35970254 -0.50966399 -1.30046302 -1.05109659 -0.20051794
[85] -1.31925090 -0.57285960 -0.30641327 -0.89225359 -0.03313499 -1.20823160 -0.80173016
[92] -1.25093133 -0.45500834 -0.95544919 -0.48746013 -1.16211589 -0.88371364 -1.32608286
[99] -1.96145487 -1.03914067 -1.43539417 -1.44564211 -2.36454033 -1.07671643 -1.36536661
[106] -1.40635836 -1.35170270 -2.35600038 -2.03660639 -3.46960938
attr(,"na.action")
[1] 15 22 26 29 40 41 42 48 60 67 73 88 90 92 93 94 97 99 107 110 115
[22] 117 122 123 131 133
attr(,"class")
[1] "omit"

> print(z_scoresGGI2010)
[1] 2.758780215 2.378147575 2.610398000 1.997514935 1.400760372 1.592689543
[7] 1.649139299 1.505595634 1.252378157 0.629817991 1.166897098 1.037869084
[13] 1.062061837 1.200766951 0.754007454 1.439468777 1.208831202 1.087867440
[19] 0.492725727 0.945936625 0.718524751 1.008837781 0.783038757 0.462081574
[25] 0.876584067 -0.055643331 0.558852584 0.629817991 1.239475356 0.658849294
[31] 0.434663121 0.658849294 0.647559343 0.228218299 0.915292471 0.462081574
[37] 0.421760319 0.062094732 0.552401183 0.318537909 0.729814702 0.386277616
[43] 0.618528040 0.405631818 0.491112877 0.247572501 0.124995888 0.621753740
[49] 0.604012389 0.276603804 0.405631818 1.084641739 -0.116931637 0.339504961
[55] 0.207251247 0.374987664 0.404018968 -0.210476947 0.302409407 0.134672989
[61] 0.200799846 0.070158982 -0.599173838 -0.336279260 0.154027191 0.065320432
[67] -0.033063428 -0.018547777 -0.015322077 -0.012096376 -0.134672989 -0.005644976
[73] 0.179832794 -0.462081574 -0.231443999 0.176607094 0.197574146 0.228218299
[79] 0.104028836 -0.145962940 0.145962940 -0.302409407 -0.105641686 -0.060481881
[85] 0.029837728 -0.302409407 0.366923414 -0.187897045 -0.274990954 -0.305635107
[91] -0.537885532 -0.428211720 -0.547562633 -1.089480290 -1.016902032 -0.494338577
[97] -1.005612081 -0.489500027 -0.421760319 -1.178187049 -0.402406117 -0.095964585
[103] -0.626592291 -0.610463789 -0.715299050 -0.916905322 -0.794328709 -0.883035468
[109] -1.171735648 -0.754007454 -0.854004165 -1.236249655 -1.189477000 -1.466887229
[115] -1.131414394 -1.347536317 -1.131414394 -1.183025600 -1.429791676 -1.720104706
[121] -1.729781807 -1.783005863 -1.642687898 -1.374954770 -1.765264511 -1.021740583
[127] -1.386244721 -2.347503422 -2.129768649 -3.520045496
attr(,"na.action")
[1] 22 41 42 60 92 93
attr(,"class")
[1] "omit"

```

9. What test would you use to compare the global average GII for 2006 to the global average GII for 2013 and why?

I would use a paired t-test and set the alternative to less

```
t.test(GGI2006,GGI2013, paired = TRUE, var.equal = FALSE, alternative = "less", conf.level = .95)
```

10. Set up this test in words and mathematically as shown in class.

A paired t-test is appropriate since I am comparing the GII scores for the same set of countries between two different years, such as comparing the GII scores for the year 2006 with the scores for the year 2013. In this case, each country's GII score is paired across the two years, making it a paired design.

Setting the alternative hypothesis to "less" indicates that I'm testing whether the mean GII score for 2006 is less than the mean GII score for 2013. This makes sense because the hypothesis is that gender inequality has decreased over time, leading to lower GII scores in 2013 compared to 2006.

11. Run this test. What are your results? Do your confidence intervals for these two years overlap?

Paired t-test

```
data:  GGI2006 and GGI2013
t = -10.87, df = 109, p-value < 2.2e-16
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
 -Inf -0.02332131
sample estimates:
mean difference
 -0.02752182
```

There is no overlap between the confidence intervals for 2006 and 2013.